

83AM01

SomeonecoolLovesMaths

May 31, 2026

Problem

On a given circle, six points $A, B, C, D, E,$ and F are chosen at random, independently and uniformly with respect to arc length. Determine the probability that the two triangles ABC and DEF are disjoint, i.e., have no common points.

Solution. It is equivalent on observing the following:

Given any 6 distinct points on a circle, how many ways can you name the points using the labels A, B, C, D, E, F such that the triangles $\triangle ABC$ and $\triangle DEF$ are disjoint.

Notice that since A, B, C must be consecutive, the favourable ways is $(6 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1) = 216$.

The total number of ways to label them is of course $6!$.

Hence the probability is $\frac{216}{6!} = \boxed{\frac{3}{10}}$.